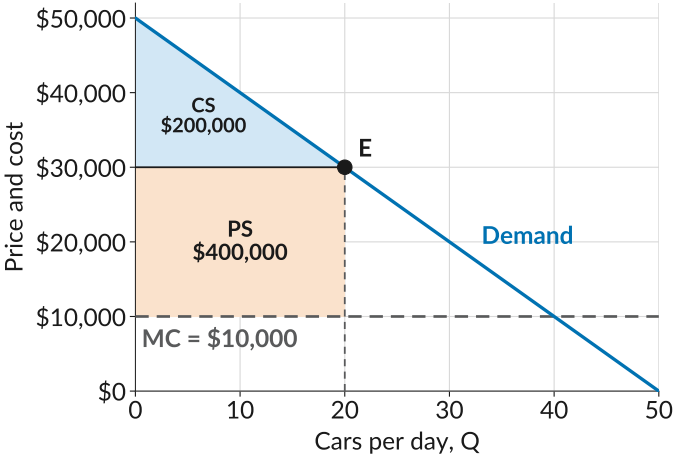
Pareto Efficiency and Deadweight Loss

ECON 201: Principles of Microeconomics
Lecture 11

Div Bhagia

Last Class: Surplus in the Market for Beautiful Cars

- The 20 sales create $600,000 of surplus a day: $200,000 for buyers and $400,000 for the firm.
- How do we judge if this is a good market outcome?
- One criterion economists use is **Pareto efficiency**.



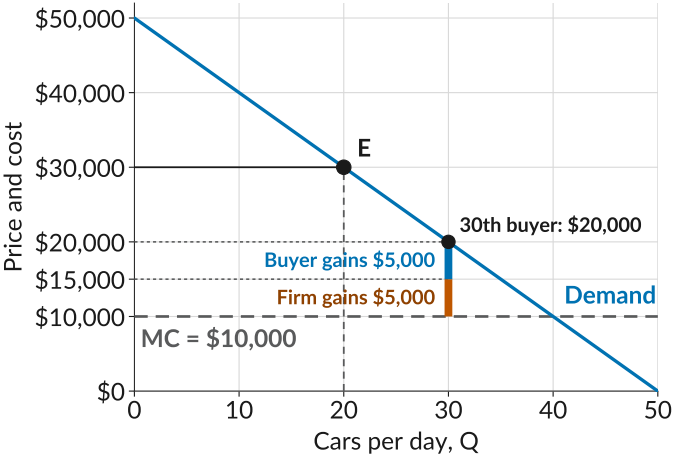
Pareto Efficiency

- An outcome is **Pareto efficient** if there is no other (feasible) outcome in which at least one person would be better off, and nobody worse off.
- We say there is a possible **Pareto improvement** if some change would make at least one person better off without making anyone worse off.
- Example: say you order pizza with your friends, and one friend eats all of it. Is that Pareto efficient?

In the market for Beautiful Cars, could there be a Pareto improvement? There could be if there are buyers who value a car at more than the $10,000 it costs to make, but do not get one.

Buyers Left Out

- The 30th buyer's WTP is $20,000, above the $10,000 MC.
- At $30,000, this buyer does not buy.
- Sell this buyer one car at $15,000: each side gains $5,000.
- Nobody is worse off, so this is a Pareto improvement.



Why Does Beautiful Cars Stop at 20?

In fact, buyers 21 to 40 all value a car above its $10,000 MC but do not get one, so the outcome at *E* is not Pareto efficient. What is stopping Beautiful Cars from selling to them?

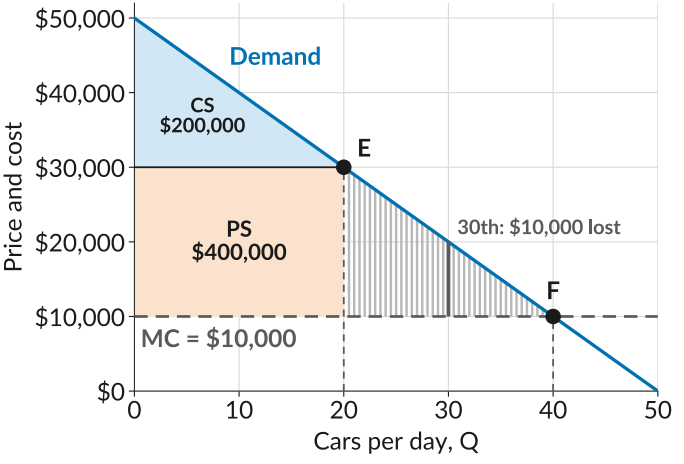
- It sets a single price for all buyers. Selling to each buyer at a different price is hard to arrange, and it is hard to know each buyer's willingness to pay.
- To reach the 30th buyer, it would have to charge $20,000 to all 30 buyers, which would lower its profit.
- Setting different prices for different buyers, based on their willingness to pay, is called **price discrimination**. Where do we see this around us?

Can we measure the surplus lost from these missing sales?

Deadweight Loss

- Buyers 21 to 40 value a car above its MC but do not get one, so each loses WTP − MC.
- **Deadweight loss (DWL)**: the total surplus lost.
- Calculate using the area of a triangle:

$$\text{DWL} = \tfrac{1}{2} \times 20 \times 20{,}000$$
$$= 200{,}000$$



The Deadweight Loss of Christmas

- Economist Joel Waldfogel calculated the deadweight loss of Christmas.
- He asked college students what each gift cost, and what they would have paid for it.
- Gifts were worth 10% to a third less than they cost: a $4 to $13 billion deadweight loss in the US in 1992.
- A dollar from friends was worth about 99 cents; from grandparents, about 63 cents (though grandparents seem to know it: 43% of their gifts were cash).



Source: Waldfogel (1993), *American Economic Review* 83(5).

What to Do Next

- Before next class: read section **7.7**, and go over the **slides** and the **worksheet**.
- Section 7.7 also asks what happens if the firm could charge every buyer their full willingness to pay (Exercise 7.3). We skip that exercise for now.
- Work through the **practice problems** for this lecture.
- Next week we look at market power: how firms stand out from the crowd, and why some markets end up with a single firm.
- **Quiz 5 is Monday, October 5**, and covers Lectures 10 and 11.